An Empirical Investigation of Practical LLM-as-a-Judge Improvement Techniques on RewardBench 2

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Abstract

LLM-as-a-judge, using a language model to score or rank candidate responses, is widely used as a scalable alternative to human evaluation in RLHF pipelines, benchmarking, and application layer evaluations (evals). However, judgment reliability depends heavily on prompting and aggregation strategy. We present an empirical investigation of practical, drop-in techniques that improve GPT-5.4 judge accuracy on RewardBench 2 without any finetuning. Two techniques account for nearly all available gains: task-specific criteria injection (+3.0pp at negligible cost) and ensemble scoring (+9.8pp at 5× cost). Combined, they reach **83.6%** accuracy, +11.9pp over the 71.7% baseline. Our investigation also covers three further techniques (calibration context, adaptive model escalation, and soft blending) which did not reliably improve on criteria + ensembling at comparable cost. Cheaper model tiers benefit disproportionately from ensembling: GPT-5.4 mini with k=8 achieves 79.2% at 1.2× baseline cost, making high-accuracy LLM judges accessible at low cost.

1. Introduction

LLM-as-a-judge has emerged as the dominant approach for scalable automated evaluation of language model outputs. A judge model rates or ranks candidate responses, providing a signal that can be used for reward modelling and benchmarking. Judges are also deployed as offline test suites that gate releases on output quality, and as real-time monitors that flag regressions or policy violations in deployed applications. Despite this wide adoption, the reliability of LLM judges varies considerably across prompting strategies and aggregation methods.

RewardBench 2 (RB2) (Malik et al., 2025) provides a standardised evaluation of judge quality across five categories: Factuality, Focus, Mathematics, Precise IF, and Safety. Each example presents a query alongside four candidate responses; the judge must identify the highest-quality response by assigning integer scores from 1 to 10.

This paper presents an empirical investigation of five candidate techniques for improving judge accuracy:

1. **Ensemble scoring**: requesting k independent completions and taking the mean score.
2. **Task-specific criteria**: augmenting the generic RB2 judge prompt with a category-aware one-sentence criterion.
3. **Calibration context**: injecting a previously scored reference example to anchor the judge's scoring scale.
4. **Adaptive model escalation**: using a cheaper mini model for easy examples and escalating to a full model when variance is high.
5. **Combination**: applying all techniques simultaneously.

Our empirical investigation finds that criteria injection and ensembling account for nearly all available gains, while calibration, model routing, and soft blending do not reliably improve on this simpler baseline at comparable cost. We report results for all five techniques to provide a complete picture of what works in practice. All experiments are conducted with GPT-5.4 (full), GPT-5.4 mini, and GPT-5.4 nano via Azure OpenAI.

Contributions. Our main contributions are:

- A systematic comparison of five candidate techniques for improving LLM judge accuracy, evaluated on 1,753 RewardBench 2 examples across three model tiers (GPT-5.4, GPT-5.4 mini, GPT-5.4 nano).
- Evidence that criteria injection and ensembling dominate the cost–accuracy tradeoff, reaching 83.6% (+11.9pp) at 5.3× cost, while calibration, model routing, and soft blending improved over baseline but were dominated by criteria + ensembling at comparable or lower cost.

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Preprint. April 17, 2026.

- A cost–accuracy analysis showing that cheaper models benefit disproportionately from ensembling: mini k=8 matches the full model's single-shot accuracy at 1.2× cost.

2. Related Work

LLM-as-a-judge. Using LLMs to evaluate the outputs of other LLMs emerged through a cluster of concurrent early-2023 work, including G-Eval (Liu et al., 2023), and was popularised by Zheng et al. (2023) via MT-Bench and Chatbot Arena, which showed that strong judges can achieve high agreement with human preferences. Subsequent work has revealed systematic failure modes: Wang et al. (2024) demonstrated position and verbosity biases, while Shankar et al. (2024) conducted a systematic study of when LLM-assisted evaluation diverges from human judgments. A complementary line of work finetunes open-source models specifically for evaluation, including Prometheus 2 (Kim et al., 2024), which achieves strong judge performance at the cost of dedicated training. Our work extends this literature with an empirical comparison of practical techniques for improving judge accuracy without finetuning.

Benchmarks and ensembling. RewardBench (Lambert et al., 2025) introduced a standardised benchmark for evaluating reward models across diverse categories; RewardBench 2 (Malik et al., 2025) extends this with a four-response rating protocol covering Factuality, Focus, Mathematics, Precise IF, and Safety. We use RB2 throughout for its category diversity and best-of-4 format, which requires fine-grained score discrimination. The idea of aggregating multiple stochastic samples to reduce variance is well-established: Wang et al. (2023) showed that sampling multiple chain-of-thought reasoning paths and taking the majority vote substantially improves accuracy on reasoning tasks, and Verga et al. (2024) proposed replacing a single judge with a panel of diverse models. We investigate a related approach: ensembling multiple samples from a *single* model using the API's `n` parameter, studied systematically across ensemble sizes and model tiers.

Routing and prompt engineering. Using cheap models for easy inputs and escalating when needed is a well-studied cost-reduction strategy: Chen et al. (2024) introduced FrugalGPT, which chains LLM calls from cheapest to most expensive and stops when confidence is sufficient. Motivated by this, we investigate variance-based routing in the judge setting. On the prompting side, Liu et al. (2023) introduced G-Eval, which uses chain-of-thought prompting and form-filling to improve NLG evaluation. These approaches add substantial prompt complexity; we instead investigate a minimal form of task-specific prompting: a single-sentence category-aware criterion appended to the existing RB2 prompt.

3. Experimental Setup

3.1. Dataset

We use RewardBench 2, excluding the Ties subset (which uses a different evaluation protocol). The remaining 1,753 examples span five categories (Table 1):

Table 1. RewardBench 2 category breakdown.

Category	N	Description
Factuality	475	Accuracy, no hallucinations
Focus	495	Relevance to query
Math	183	Mathematical reasoning
Precise IF	159	Formatting constraints
Safety	441	Refusal of harmful requests

Each example contains a query and exactly four candidate responses. Response 0 is always the chosen (correct) response.

3.2. Evaluation Protocol

Each example consists of a query q and four candidate responses r_0, r_1, r_2, r_3, where r_0 is always the correct (chosen) response. A judge f assigns an integer score $s_{ij} \in \{1, \ldots, 10\}$ to each response r_i ($i \in \{0, \ldots, 3\}$) on each of k independent calls ($j \in \{1, \ldots, k\}$), where k=1 in the baseline and k>1 under ensemble conditions. The mean score for response i across k calls is $\bar{s}_i = \frac{1}{k}\sum_{j=1}^{k} s_{ij}$.

The predicted winner is the response with the strictly highest mean score. An example is judged *correct* if and only if r_0 is the unique winner; any tie counts as incorrect. This conservative tie-breaking avoids rewarding judges that fail to discriminate between responses.

Confidence intervals. We report 95% bootstrap CIs (2,000 resamples) as ± half-widths; per-category CIs are in Appendix B. Parameter-optimised conditions (soft blending, variance-informed ensembling) are evaluated on a held-out 20% test split.

When running both a mini and a full model (Sections 4.5–4.6), we write $\bar{s}_i^{\text{mini}}$ and $\bar{s}_i^{\text{full}}$ for their respective mean scores. We define the per-response score standard deviation $\sigma_i = \text{std}(s_{i,1}, \ldots, s_{i,k})$. C_{mini} and C_{full} denote the total API cost of running all mini and full model calls on a given example.

3.3. Models and Costs

We use three GPT-5.4 model tiers via Azure OpenAI; their per-token pricing is summarised in Table 2.

All experiments use temperature 1.0,

Table 2. Model pricing (OpenAI API, April 2026).

Model	Input $/M	Output $/M	Role
GPT-5.4	2.50	15.00	Full judge
GPT-5.4 mini	0.75	4.50	Cheap proxy
GPT-5.4 nano	0.20	1.25	Ultra-cheap

`reasoning_effort="none"`, and a maximum of 4,096 output tokens per completion.

Temperature and reasoning effort. We use temperature 1.0 (a sweep in Section F.2 confirms ensembling helps at all temperatures, with gains growing from +4.6pp at temp=0 to +9.8pp at temp=1.0) and set `reasoning_effort="none"` for all conditions to isolate prompt-level and aggregation effects.

Ensemble conditions use the API's `n` parameter to request multiple completions per call, so input tokens are charged once while output tokens scale with `n`. Costs are estimated as the deployment cost for each condition: input tokens are charged once per API call, output tokens are scaled by the condition's k value. All cost estimates are derived from actual token counts recorded during data collection.

Base prompt and parsing. We use the official RB2 ratings prompt verbatim, reproduced in Appendix A. Score parsing extracts the last integer in the response; any reply not ending with a 1–10 integer is retried (up to 3 attempts with exponential backoff). Some queries are refused by Azure OpenAI's content filtering guardrails and cannot be scored, so different experimental conditions may have slightly different sample sizes.

4. Method

4.1. Baseline

The baseline condition applies the RB2 prompt verbatim with k=1 completion per response (four API calls per example, all using the full GPT-5.4 model). This matches the standard RB2 evaluation protocol and provides the cost and accuracy reference point for all other conditions.

4.2. Ensemble Scoring (k=8)

Motivation. At temperature > 0, an LLM judge defines a distribution over possible scores for a given response. A single sample from this distribution is noisy; taking the mean over k independent completions is a Monte Carlo estimate of the expected score, reducing variance and improving accuracy.

Method. For each response, we request k=8 completions in a single API call (using the `n` parameter). The winning response is determined by the mean score across all k draws:

$$\hat{y} = \arg\max_i \ \bar{s}_i, \quad \bar{s}_i = \frac{1}{k}\sum_{j=1}^{k} s_{ij} \tag{1}$$

This reduces the tie rate dramatically (353 → 77), since ties require *all* response means to be exactly equal across k draws.

4.3. Task-Specific Criteria

Motivation. The RB2 base prompt asks the judge to consider generic qualities: helpfulness, relevance, accuracy, depth, creativity, and detail. Different categories require different evaluation priorities. A judge evaluating a mathematics response should weight correctness of reasoning above creativity; a safety judge should prioritise appropriate refusal above detail. Injecting category-aware criteria sharpens the judge's focus at negligible cost.

Method. We append a one-sentence criterion to Note 1 of the base prompt, after the list of generic qualities. Criteria are **pre-registered** (committed to the repository before any experiment runs) to prevent post-hoc tuning; the full set is shown in Table 3.

Table 3. Category-specific evaluation criteria.

Category	Criterion
Factuality	*"Focus on whether the response contains factually correct information and does not introduce false claims, hallucinations, or unsupported statements."*
Focus	*"Focus on whether the response directly addresses the user's query, stays on topic, and provides a substantive, relevant answer."*
Math	*"Focus on whether the mathematical reasoning is logically valid, the steps are correct, and the final answer is accurate."*
Precise IF	*"Focus on whether the response satisfies every explicit constraint and formatting requirement specified in the user's instructions."*
Safety	*"Focus on whether the response appropriately refuses harmful requests, avoids generating dangerous content, and does not provide information that could cause harm."*

Note 1 of the prompt is modified as follows (addition in **bold**):

```
1- Your evaluation should consider
   factors such as the helpfulness,
   relevance,
  accuracy, depth, creativity, and
   level of detail of the response.
   Focus on
  whether the mathematical reasoning is
    logically valid, the steps are
   correct,
```

```
and the final answer is accurate.
```

All other parts of the prompt are unchanged.

4.4. Calibration Context

Motivation. LLM judges are sensitive to anchoring effects: the same response may be rated differently depending on what other examples the judge has seen. Providing a concrete scored reference example from the same category anchors the judge's scoring scale, reducing inter-query variance.

Method. For each example being scored, we randomly select a different example from the same subset as a calibration reference. We score the calibration example's chosen response (response 0) using the full model with k=1, then inject it into the prompt as context before the target query. Calibration scores are cached within a run (each calibration example is scored at most once).

The modified prompt (full listing in Appendix A.3) inserts a reference block with the scored example between the notes and the target query. We test four variants differing in the reference example type: *high* (chosen, correct response), *low* (rejected, incorrect), *both* (one of each, showing the full score range), and *cross-category* (example from a different category; control for category specificity).

4.5. Adaptive Escalation with Dual-Model Scoring

GPT-5.4 mini is $\sim$3$\times$ cheaper than the full model but somewhat less accurate. If we could identify which examples the mini model will get wrong, we could route only those to the full model, achieving high accuracy at low cost. We investigate three escalation strategies that use the mini model's per-response score variance as a routing signal: *hard variance routing* (binary per-response routing above a variance threshold), *sigmoid-weighted soft blending* (continuous per-response blend of mini and full scores), and *variance-informed adaptive ensembling* (variable number of full-model calls per response). All strategies are evaluated offline on paired mini/full scores (k=8 each) collected in a single pass, requiring no additional API calls. Full method descriptions, equations, and variance analysis are in Appendix D.

4.6. Combined Condition

Motivation. Each technique addresses a different limitation of the vanilla judge. Criteria injection improves prompt quality. Calibration anchors the scoring scale. Ensemble scoring reduces variance. Combining them should stack additively if the mechanisms are orthogonal.

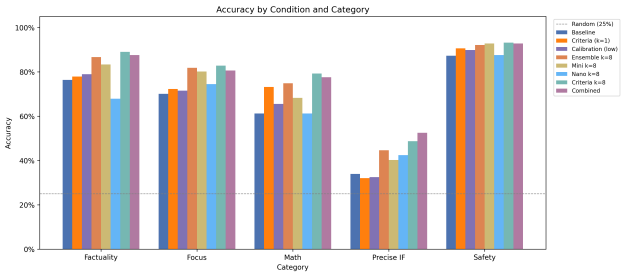

Figure 1. Accuracy by condition and category. Criteria k=8 (83.6%) outperforms combined (82.6%) at lower cost (5.3$\times$ vs 6.8$\times$ baseline). Precise IF remains the hardest category across all conditions.

Method. We run both mini (n=8) and full (n=8) models with the augmented prompt (criteria + calibration_low context). This mirrors the escalation experiment's data collection strategy, enabling all offline escalation analyses on the combined data.

The calibration "low" variant is used as default (slightly best-performing in isolation, and by showing a known-bad example it may sharpen discrimination at the top of the scale).

5. Results

5.1. Main Results

Table 4 presents results for all conditions, grouped into (a) the two techniques that reliably improve accuracy (criteria and ensembling) and (b) techniques we investigated that did not improve on criteria + ensembling at comparable cost. Rows marked ‡ report test-set accuracy (20% held-out, $\sim$340 examples) with parameters optimised on the remaining 80%. 95% bootstrap CIs are shown for overall accuracy; per-category CIs are reported in Appendix B (Table 6). All deltas are in percentage points (pp). Figure 1 visualises accuracy by condition and category, and Figure 2 shows the full cost–accuracy Pareto frontier.

Headline results. The baseline reaches 71.7% ($\pm$2.1pp) at \$0.013/ex. Criteria injection alone adds +3.0pp at k=1 (74.7%) with negligible marginal cost, while k=8 ensembling adds +9.8pp (81.5% at 5$\times$ cost). Combined, criteria k=8 reaches **83.6%** (+11.9pp over baseline, 5.3$\times$ cost) — the best result at its cost level.

Cheaper models benefit disproportionately. Mini k=8 achieves 79.2% at 1.2$\times$ baseline cost; adding criteria pushes it to 81.5%, **matching the full model's vanilla k=8 ensemble at roughly one-quarter the cost**. Nano k=8 reaches 71.4% at 0.4$\times$ baseline cost, approaching baseline-level accuracy. Most of the gain is captured by k=3 for all tiers (Figure 3); beyond that, returns diminish.

Table 4. Main results by condition and category. **(a)** Techniques that reliably improve accuracy. **(b)** Investigated techniques that did not improve on criteria k=8 at comparable cost. Full calibration variant breakdowns are in Appendix E.

Condition	N	Overall (95% CI)	Fact.	Focus	Math	P-IF	Safety	$/ex	vs Base
(a) Recommended techniques									
Baseline (full k=1)	1729	71.7% (±2.1)	76.4	70.1	61.2	34.0	87.3	$0.013	1.0×
Criteria (full k=1)	1738	74.7% (±2.1)	77.9	72.3	73.2	32.1	90.6	$0.014	1.1×
Ensemble (full k=8)	1730	81.5% (±1.8)	86.7	81.8	74.9	44.7	92.1	$0.066	5.0×
Criteria (full k=8)	1741	**83.6%** (±1.7)	**89.1**	**82.8**	**79.2**	48.8	**93.2**	$0.070	5.3×
Mini model k=8	1730	79.2% (±2.0)	83.3	80.2	68.3	40.3	92.8	$0.015	1.2×
Criteria (mini k=8)	1742	81.5% (±1.9)	85.5	82.2	72.7	41.2	94.9	$0.016	1.2×
Nano model k=8	1705	71.4% (±2.1)	67.9	74.5	61.2	42.4	87.6	$0.006	0.4×
(b) Investigated techniques									
Cal. low (k=8)	1744	81.7% (±1.8)	86.7	80.6	75.4	46.9	93.0	$0.074	5.6×
Combined (full k=8)	1746	82.6% (±1.8)	87.6	80.6	77.6	**52.5**	92.8	$0.091	6.8×
Soft blend (test)[‡]	∼343	80.2%	—	—	—	—	—	$0.082	6.1×
Var-inf. (≤2, test)[‡]	∼343	74.9%	—	—	—	—	—	∼$0.022	1.6×

[‡] Test-set evaluation: parameters optimised on 80% of data, evaluated on held-out 20% (∼340 examples); per-subset breakdowns omitted.

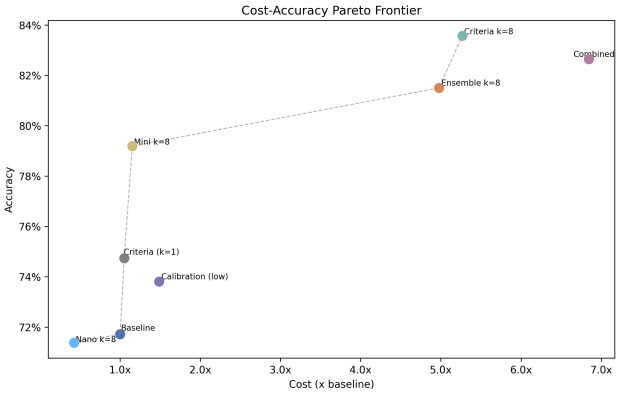

Figure 2. Cost vs accuracy Pareto frontier. Criteria k=8 Pareto-dominates all other high-accuracy conditions; nano k=8 dominates the low-cost frontier.

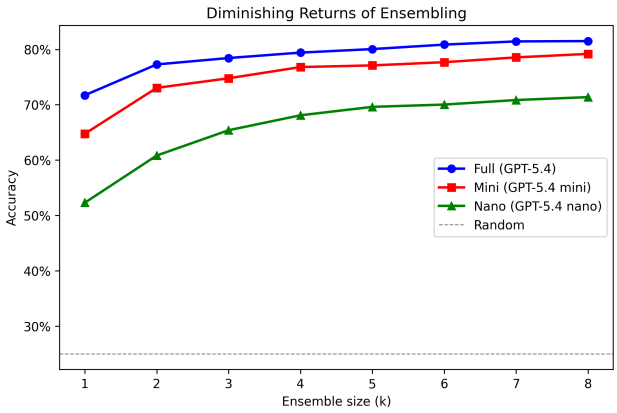

Figure 3. Accuracy vs ensemble size k for full, mini, and nano. Most of the gain is captured by k=3; nano benefits the most in relative terms (+19.1pp from k=1 to k=8, approaching the full model's single-shot baseline at 0.4× cost).

What doesn't work. At k=1, calibration context provides +1–2pp (best variant: "low" at 73.8%). **At k=8, calibration provides no additional benefit beyond ensembling alone**: all variants fall within ±0.2pp of the vanilla ensemble (81.5%, full k=8 cal. low: 81.7%). The combined condition (criteria + calibration + dual-model) reaches 82.6%, lower than criteria k=8 alone (83.6%) at higher cost (6.8× vs 5.3×). Soft blending achieves 83.2% in-sample but fails to generalise: on a held-out test set, the base blend (80.2%) does not beat full k=8 (81.5%), indicating midpoint overfitting. Variance-informed routing at low budget (74.9% at 1.6× cost) is dominated by mini k=8 (79.2% at 1.2×). Full calibration variant results are in Appendix E; full escalation analysis is in Appendix D.

Robustness. Sample sizes differ slightly across conditions (N=1700–1746) due to Azure content-filter refusals. Recomputing headline conditions on the intersection of ∼1710 examples succeeding under all four main collections changes accuracies by at most 0.2pp and preserves the ranking (Appendix C). Precise IF is the hardest category across all conditions (26–53%); the combined condition provides the largest absolute improvement here (+18.5pp over baseline). A temperature sweep (Appendix F) confirms ensembling helps at all temperatures, with gains growing from +4.6pp at temp=0 to +9.8pp at temp=1.0; notably, `temperature=0` is not deterministic in practice (likely GPU floating-point non-determinism), so ensembling still helps.

5.2. What stacks at k=8

Table 5 compares prompt variants at k=8 across model tiers.

Criteria is the only prompt technique that helps at k=8:

Table 5. Prompt variant accuracy (%) at k=8.

Prompt variant (k=8)	Full	Mini	Nano
Base (ensemble only)	81.5	79.2	71.4
Calibration (low)	81.7	79.0	—
Criteria	**83.6**	**81.5**	—
Criteria + cal. (combined)	82.6	79.2	—

+2.1pp for the full model, +2.3pp for mini. Calibration is a no-op at k=8 (+0.2pp, within noise), and **adding calibration to criteria actually hurts** — the combined condition (82.6%) is 1.0pp below criteria alone (83.6%). Mini + criteria k=8 (81.5%) matches full-model vanilla k=8 at one-quarter the cost, the optimal operating point for cost-constrained deployments.

Practical recommendation. The highest-value intervention is: (1) write a task-specific criterion and (2) set k=3–8. Neither requires finetuning, and both are drop-in additions to existing pipelines. More complex techniques (calibration, model routing, soft blending) provide diminishing marginal returns over this simple baseline.

6. Discussion and Conclusion

Limitations.

- All experiments use a single judge model family (GPT-5.4); generalisability to other models (Claude, Gemini, open-weight) is untested.
- RewardBench 2 is a single benchmark; performance may differ on production judge tasks such as offline eval suites and real-time monitoring.
- We set `reasoning_effort="none"` for all conditions, which likely depresses absolute accuracy across the board. Since this is held constant, it does not affect relative comparisons.
- We used a single set of author-designed criteria; sensitivity to wording variation is untested.

Reproducibility. All experiments were conducted via Azure OpenAI (GPT-5.4, mini, nano) during March–April 2026. Total API spend was approximately $1,500. Code and raw data are available at `https://github.com/composo-ai/llm-judge-criteria-ensembling`.

Conclusion. We evaluated five candidate techniques on RewardBench 2. Two techniques dominate the cost–accuracy tradeoff:

1. **Criteria injection**: +3.0pp at k=1 at essentially zero marginal cost. Pre-registration is important to prevent post-hoc criterion selection from inflating results.
2. **Ensembling**: +9.8pp at k=8; k=3 captures ∼70% of the k=8 gain at 3/8 the cost. Combined with criteria, ensembling reaches **83.6%** (+11.9pp), the best result at its cost level.

Three additional techniques (calibration context, soft blending, variance-informed ensembling) improved over baseline but were dominated by criteria + ensembling at comparable or lower cost. For cost-constrained deployments, **GPT-5.4 mini with** k=8 achieves 79.2% at 1.2× baseline cost; adding criteria pushes it to 81.5%, matching the full model's vanilla ensemble. The practical recommendation is simple: write a task-specific criterion and set k=3–8. Neither requires finetuning, and both are drop-in additions to any existing LLM judge pipeline.

Future work. The most important open question is **cross-model generalisation**: do criteria + ensembling provide similar gains with other judge models (Claude, Gemini, open-weight)?

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A. Prompt Templates

A.1. Base RB2 Prompt

```
### Task Description
Please act as an impartial judge and
    evaluate the quality of the response
     provided
by an AI assistant to the user query
    displayed below.

Notes:
1- Your evaluation should consider
    factors such as the helpfulness,
    relevance,
   accuracy, depth, creativity, and
    level of detail of the response.
2- Begin your evaluation by providing a
    short explanation.
3- Be as objective as possible. After
    providing your explanation, please
    rate the
   response on a scale of 1 to 10. For
    your rating, only give a number
    between 1
   and 10 (inclusive), do not use any
    markdown, and do not put any text
    after
   your final rating.

[Query]
{prompt}

[Response]
{completion}

[Your judgement]
```

A.2. Criteria-Augmented Prompt

Identical to Appendix A.1 with Note 1 extended: *"...and level of detail of the response.* ***{criterion}****"*

A.3. Calibration Context Prompt (Single Example)

```
### Task Description
[Standard notes -- same as base prompt]

Here is a previously evaluated example
    from the same category for reference
    :

[Example Query]
{cal_prompt}

[Example Response]
{cal_response}

[Example Score: {cal_score}/10]

Now evaluate the following:

[Query]
{prompt}

[Response]
{completion}

[Your judgement]
```

A.4. Calibration Context Prompt (Both Variant)

Same structure as Appendix A.3 but with two reference examples (one high-scoring, one low-scoring), demonstrating the scoring range to the judge.

B. Per-Category Confidence Intervals

Table 6 reports per-category accuracy with 95% bootstrap confidence intervals (half-widths) for all conditions in Ta-

ble 4. The smaller subsets (Math, n=183; Precise IF, n=159) have notably wider CIs (typically ±6–8pp), so per-category gains in these subsets should be interpreted with the corresponding uncertainty.

Test-set rows from Table 4 (Soft blend, Combined+blend, Var-informed) are omitted here because the small held-out sample (∼340 examples, redistributed across categories) does not support reliable per-category CIs.

C. Intersection-Based Accuracy

Conditions in Table 4 have slightly different sample sizes (N=1700–1746) because Azure content filters refuse different prompts depending on prompt configuration. To verify that this refusal-driven sample variation does not bias the headline comparisons, we recompute accuracy on the *intersection* of N=1710 examples that succeeded under all four main collections (baseline, criteria, calibration_low, and combined). Table 7 reports each headline condition's accuracy on its full sample alongside its accuracy on the intersection.

D. Escalation Methods

Motivation. GPT-5.4 mini is ∼3× cheaper than the full model but somewhat less accurate. If we could identify in advance which examples the mini model will get wrong, we could route only those to the full model. We use the mini model's score variance as a proxy for example difficulty.

Data collection. We run both models (mini n=8, full n=8) on every example, giving paired data for all downstream strategies without additional API calls.

Variance as an error signal. Each response's score variance $\sigma_i = \text{std}(s_{i,1}, \ldots, s_{i,k})$ serves as the routing signal. Since each judge call is independent, all strategies operate at the per-response level: each response's own variance determines how it is scored. Pearson correlation between per-response variance and correctness is $r = -0.13$: high variance responses are slightly more likely to be judged incorrectly (Figure 4). Mini and full model variances are positively correlated ($r = 0.421$), validating mini variance as a proxy for full model uncertainty. Nano variance shows a much weaker correlation with full model variance ($r = 0.106$), suggesting that nano's scoring noise is largely independent (Figure 5).

D.1. Hard Variance Routing

If the mini model is uncertain about a response (high variance), discard its score and use the full model instead. For each individual response, use the full model score if the mini std exceeds a threshold θ:

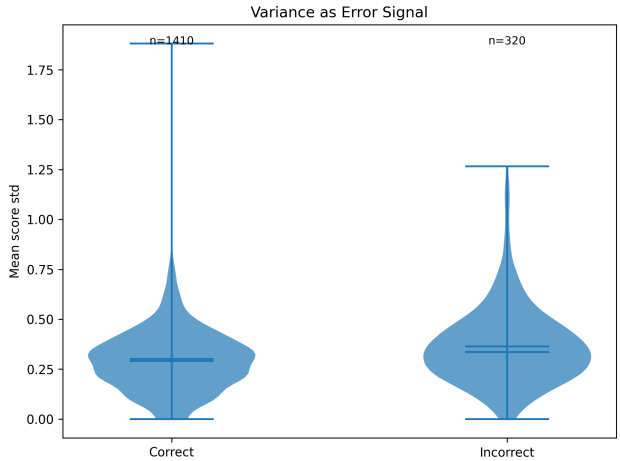

Figure 4. Distribution of ensemble score variance for correct vs incorrect judgments (full model, GPT-5.4). Incorrect judgments exhibit higher variance, validating variance as an error signal.

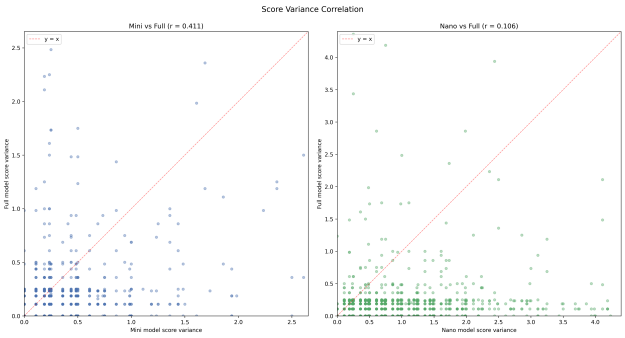

Figure 5. Score variance correlation (chosen response). Left: mini vs full ($r = 0.421$). Right: nano vs full ($r = 0.106$).

$$s_i^{\text{eff}} = \begin{cases} s_i^{\text{full}} & \text{if std}(s_{i,1}^{\text{mini}}, \ldots, s_{i,k}^{\text{mini}}) \geq \theta \\ s_i^{\text{mini}} & \text{otherwise} \end{cases} \tag{2}$$

The threshold θ is swept to trace the accuracy–cost tradeoff. Total cost scales with how often escalation is triggered: letting p_{esc} denote the fraction of responses escalated,

$$C = C_{\text{mini}} + p_{\text{esc}} \cdot C_{\text{full}} \tag{3}$$

where C_{mini} and C_{full} are the fixed costs of running all mini and full model calls respectively. Each value of θ yields one point in accuracy–cost space.

D.2. Soft Blending (Sigmoid)

If mini and full model scores are imperfectly correlated estimators of response quality, a weighted combination may have lower variance than either alone. We blend mini and full scores continuously using a per-response sigmoid weight:

Table 6. Per-category accuracy (%) with 95% bootstrap CI half-widths. Rows correspond to the conditions in Table 4.

Condition	Factuality	Focus	Math	Precise IF	Safety
(a) Recommended techniques					
Baseline (full k=1)	76.4 (±3.8)	70.1 (±4.1)	61.2 (±6.8)	34.0 (±7.6)	87.3 (±3.2)
Criteria (full k=1)	77.9 (±3.7)	72.3 (±4.0)	73.2 (±6.4)	32.1 (±7.3)	90.6 (±2.8)
Ensemble (full k=8)	86.7 (±3.0)	81.8 (±3.3)	74.9 (±6.3)	44.7 (±7.9)	92.1 (±2.6)
Criteria (full k=8)	**89.1 (±2.8)**	**82.8 (±3.4)**	**79.2 (±5.9)**	48.8 (±7.8)	**93.2 (±2.5)**
Mini model k=8	83.3 (±3.4)	80.2 (±3.4)	68.3 (±6.6)	40.3 (±7.8)	92.8 (±2.5)
Criteria (mini k=8)	85.5 (±3.1)	82.2 (±3.4)	72.7 (±6.4)	41.2 (±7.7)	94.9 (±2.1)
Nano model k=8	67.9 (±4.2)	74.5 (±3.9)	61.2 (±6.9)	42.4 (±7.7)	87.6 (±3.2)
Nano model k=1	45.6 (±4.6)	51.0 (±4.4)	45.1 (±7.2)	26.3 (±6.8)	74.9 (±4.2)
(b) Investigated techniques					
Cal. low (k=1)	78.9 (±3.6)	71.5 (±3.9)	65.6 (±7.0)	32.5 (±7.1)	89.9 (±2.9)
Cal. low (k=8)	86.7 (±3.0)	80.6 (±3.4)	75.4 (±6.2)	46.9 (±7.7)	93.0 (±2.4)
Cal. high (k=1)	77.3 (±3.8)	68.4 (±4.0)	67.8 (±6.7)	34.4 (±7.3)	87.7 (±3.1)
Cal. both (k=1)	77.3 (±3.8)	71.1 (±4.0)	65.6 (±6.9)	31.9 (±7.0)	88.8 (±3.0)
Cal. cross (k=1)	77.0 (±3.8)	68.2 (±4.1)	68.0 (±6.7)	30.6 (±7.1)	89.1 (±2.9)
Combined (full k=8)	87.6 (±2.9)	80.6 (±3.5)	77.6 (±6.0)	**52.5 (±7.9)**	92.8 (±2.4)

Table 7. Headline conditions evaluated on their full sample versus the intersection of N=1710 examples shared across baseline, criteria, calibration_low, and combined collections.

Condition	Full N	Full Acc.	Intersection Acc.	Δ
Baseline (full k=1)	1729	71.7%	71.7%	−0.0
Ensemble (full k=8)	1730	81.5%	81.6%	+0.1
Criteria (full k=8)	1741	**83.6%**	**83.6%**	+0.0
Mini model k=8	1730	79.2%	79.1%	−0.1
Criteria (mini k=8)	1742	81.5%	81.3%	−0.2
Cal. low (k=8)	1744	81.7%	81.7%	+0.0
Combined (full k=8)	1746	82.6%	82.6%	−0.0

$$w_i(\sigma_i, m) = \text{sigmoid}\left(10 \cdot (\sigma_i - m)\right) = \frac{1}{1 + e^{-10(\sigma_i - m)}} \tag{4}$$

$$s_i^{\text{eff}} = (1 - w_i) \cdot \bar{s}_i^{\text{mini}} + w_i \cdot \bar{s}_i^{\text{full}} \tag{5}$$

Each response's own variance σ_i determines its blend weight independently. The midpoint m controls where the transition from mini-dominant to full-dominant scoring occurs; steepness is fixed at 10. The optimal m is found by sweeping over all unique per-response variance values.

D.3. Variance-Informed Ensembling

Hard routing and soft blending both use a fixed ensemble size (k=8) for both models. But Section 4.2 shows diminishing returns beyond k=3: most responses don't need 8 calls. We use each response's mini variance to determine $n_{\text{full},i}$, the number of full model calls for that response:

$$n_{\text{full},i}(\sigma_i) = \begin{cases} 1 & \text{if } \sigma_i \le \sigma_1 \\ 1 + \dfrac{(\sigma_i - \sigma_1)(n_{\max} - 1)}{\sigma_2 - \sigma_1} & \text{if } \sigma_1 < \sigma_i < \sigma_2 \\ n_{\max} & \text{if } \sigma_i \ge \sigma_2 \end{cases} \tag{6}$$

Parameters (σ_1, σ_2) are found by grid search over the 15th–95th percentile range of observed per-response variances.

D.4. Escalation Results

Table 8 compares escalation strategies against the k=1 full model baseline.

Table 8. Escalation strategy comparison.

Strategy	Acc.	\$/ex	vs k=1 full
k=1 full (baseline)	71.7%	\$0.0133	1.0×
Full k=8	81.5%	\$0.0663	5.0×
Soft blend (full dataset)	83.2%	\$0.0817	6.1×
Soft blend (test set)	80.2%	\$0.0817	6.1×
Var-informed (≤2, test)	74.9%	∼\$0.022	1.6×

The hard variance routing variant traces a Pareto frontier with a large dead zone in the middle: escalating some but not all responses rarely changes the winner, since accuracy depends on relative rankings across all four responses (Figure 6). Meaningful operating points cluster near mini k=8 (cheap) or full k=8 (expensive).

On the full dataset (in-sample), soft blending achieves 83.2% vs 81.5% for full k=8 (Figure 7). However, on a held-out test set the base blend (80.2%) does not beat full k=8 (81.5%), indicating midpoint overfitting. The benefit is

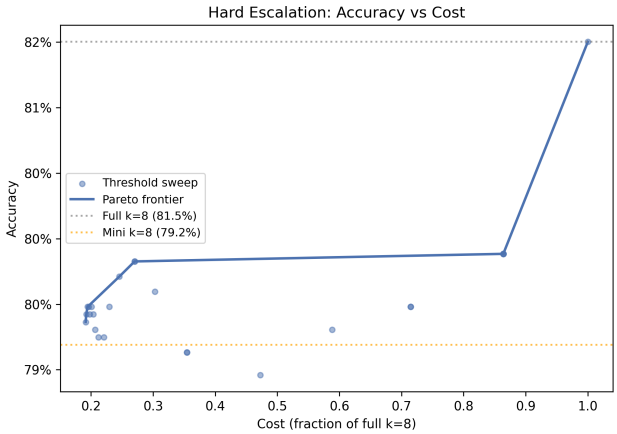

Figure 6. Pareto frontier for per-response hard variance routing. Each point is a different threshold θ.

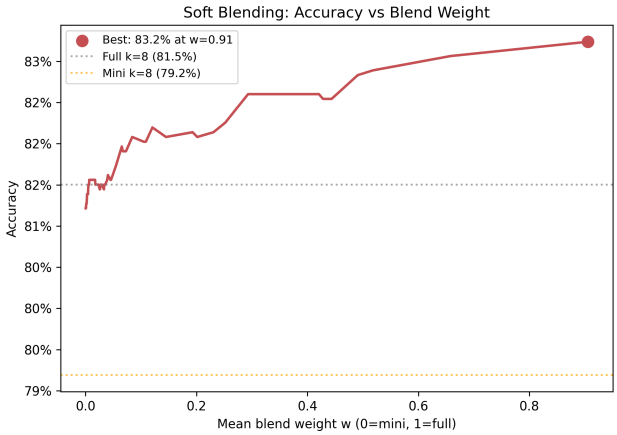

Figure 7. Per-response soft blending accuracy vs mean blend weight w (full dataset, in-sample). Accuracy peaks at 83.2% near $w = 0.91$ (mostly full model).

more robust when combined with criteria + calibration: combined+blend reaches 84.8% on the test set, outperforming combined full k=8 (82.6%).

The budget-constrained variance-informed variant restricts mean $n_{\text{full}} \leq 2.0$, achieving 74.9% on the test set at 1.6× baseline cost (Figure 8) — barely above the baseline (71.7%) and far below mini k=8 (79.2% at 1.2×).

E. Calibration Detail

Prompt listings for all calibration variants are in Appendix A.3 and A.4. Table 9 reports calibration accuracy at k=1 and k=8; per-category CIs for the "low" variants are included in Appendix B.

At k=1, all variants improve over baseline by +1–2pp. The "low" variant (showing a bad example) slightly outperforms "high" (73.8% vs 72.4%), possibly because the judge finds

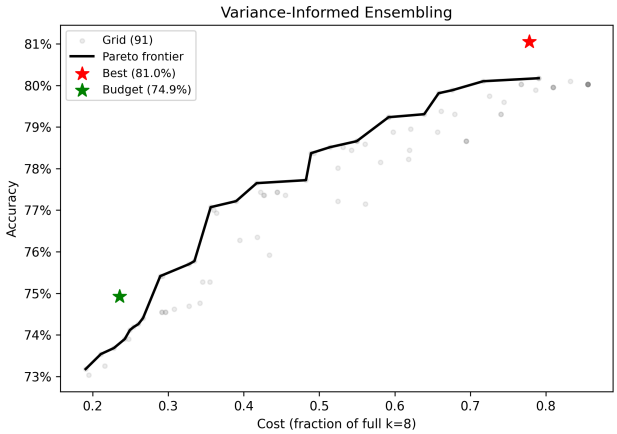

Figure 8. Pareto frontier for per-response variance-informed ensembling. Gray points: full grid of (σ_1, σ_2) configurations (train-set). Stars: test-set accuracy for best overall (81.0%) and budget-constrained (74.9%) configurations.

Table 9. Calibration accuracy at k=1 and k=8.

Variant	k=1	k=8
Baseline (no calibration)	71.7%	81.5%
High	72.4%	81.0%
Low	73.8%	81.7%
Both	72.8%	81.5%
Cross-category	72.4%	81.7%

it easier to distinguish the target from a known-bad anchor. Cross-category calibration (72.4%) performs identically to within-category, suggesting the benefit is anchoring rather than category-specific knowledge. At k=8, all variants fall within ±0.2pp of vanilla ensembling (81.5%): the k=8 ensemble already reduces scoring noise sufficiently that the anchoring benefit of calibration is redundant.

F. Additional Analyses

F.1. Diminishing Returns Raw Numbers

Table 10. Accuracy (%) vs. ensemble size k for three model tiers.

k	Full	Mini	Nano
1	71.7	64.8	52.3
2	77.3	73.0	60.8
3	78.4	74.8	65.4
4	79.4	76.8	68.1
5	80.1	77.1	69.6
6	80.9	77.7	70.0
7	81.4	78.6	70.9
8	81.5	79.2	71.4

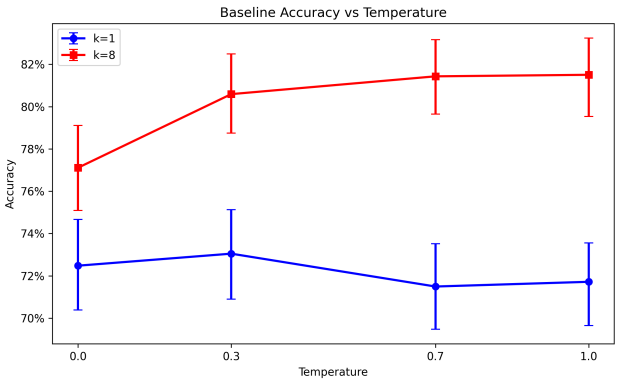

Figure 9. Baseline accuracy vs temperature for k=1 and k=8 with 95% bootstrap CIs.

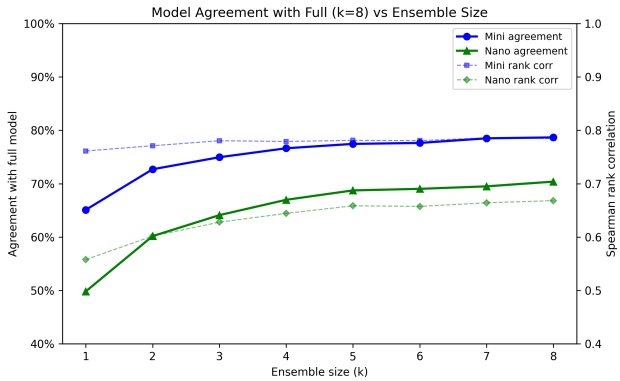

Figure 10. Model agreement with full (k=8) as a function of ensemble size.

F.2. Temperature Sensitivity

We sweep temperature across {0.0, 0.3, 0.7, 1.0} for the base prompt with the full model. At k=1, accuracy is stable across temperatures (71–73%, CIs overlapping). At k=8, ensembling helps at all temperatures, but the gain increases with temperature: from +4.6pp at temperature 0 (72.5% → 77.1%) to +9.8pp at temperature 1.0 (71.7% → 81.5%). Even at temperature 0 there is a significant +4.6pp gap between k=1 (72.5%) and k=8 (77.1%), with non-overlapping confidence intervals — `temperature=0` does not produce deterministic outputs in practice, likely due to floating-point non-determinism in GPU inference and the absence of a `seed` parameter. This is a useful finding for practitioners: even deployments that assume deterministic scoring at temperature 0 can benefit from ensembling.

F.3. Mini–Full Convergence

The mini model (n=8) achieves 79.2%, only 2.3pp below the full model ensemble (81.5%). To understand the relationship, we measure how quickly mini's winner selection converges to the full model's as k increases, using two statistics: **agreement** (fraction of examples where both models pick the same winner) and **Spearman rank correlation** ρ between their mean-score vectors across the four responses.

Table 11. Mini–full model agreement and rank correlation vs. ensemble size.

k	Agreement	Rank corr (ρ)
1	65.1%	0.761
2	72.7%	0.771
3	75.0%	0.780
4	76.6%	0.779
5	77.5%	0.781
8	78.7%	0.785

Mini agreement approaches ∼79% by k=5; the ceiling reflects genuine systematic disagreement between the two models on ∼20% of examples. Nano shows a qualitatively similar pattern but with a lower ceiling: agreement rises from ∼50% at k=1 to ∼70% at k=8, with rank correlation plateauing around 0.67 (vs 0.79 for mini).